

Auto Ticket Classification using Flow Designer

Project Report
Built on ServiceNow

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Team Size	3

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1. Introduction

1.1 Project Overview

The school IT helpdesk receives many incident requests every day from students and teachers, such as Wi-Fi issues, projector failures, password problems and slow computers. IT staff currently review each request by hand and assign a category, which is slow and inefficient.

This project automates ticket classification using **ServiceNow Flow Designer**. When a ticket is created, the flow checks keywords in the short description, fills in the Category and Subcategory automatically, and emails the caller a confirmation. The solution is entirely no-code and easy to maintain.

1.2 Purpose

- Automatically classify incidents at the time of creation.
- Reduce manual effort for IT agents.
- Improve ticket routing efficiency.
- Provide a no-code, easily maintainable solution that can scale as request volume grows.

2. Ideation Phase

2.1 Problem Statement

The school IT helpdesk handles recurring requests about network connectivity, hardware failures, account access and system performance. For every ticket, staff must identify the issue type, select the correct category and subcategory, and inform the caller. This manual process is time-consuming, error-prone and does not scale as the number of requests increases.

2.2 Empathy Map Canvas

Quadrant	Student / Teacher (Caller)	IT Staff (Agent)
Says	"My Wi-Fi is not working." "Has my request been received?"	"Too many tickets to sort by hand."
Thinks	Will anyone see my problem quickly?	Categorizing is repetitive and mistakes creep in.
Does	Raises a ticket with a short description of the issue.	Reads each ticket and picks category and subcategory.
Feels	Uncertain and frustrated while waiting.	Overloaded during busy periods.

2.3 Brainstorming

The team considered several ways to reduce manual triage and prioritized them on effort and benefit:

Idea	Effort	Benefit	Decision
Keyword-based classification in Flow Designer	Low	High	Selected
Automatic email acknowledgement to caller	Low	High	Selected
Dependent Category / Subcategory choices	Low	Medium	Selected
Machine-learning based classification	High	High	Future scope
Auto-assignment to groups and SLA tracking	Medium	High	Future scope

3. Requirement Analysis

3.1 Customer Journey Map

Stage	Caller experience	System action
1. Raise ticket	Enters caller and short description.	Record is created in the Incident WorkFlow table.
2. Classification	Nothing to do.	Flow checks keywords and sets Category and Subcategory.
3. Acknowledge	Receives a confirmation email.	Flow sends the email to the caller's address.
4. Resolution	Waits for the IT team.	Agent works on a correctly categorized ticket.

3.2 Solution Requirements

Functional requirements. The system must:

- Automatically classify IT tickets based on the issue description.
- Assign both Category and Subcategory without manual intervention.
- Support dependent choice logic between Category and Subcategory.
- Send an automated email notification to the caller upon ticket creation.
- Store ticket information in a structured and standardized format.
- Ensure easy maintenance and future scalability.

Data fields of the custom table **Incident WorkFlow** (u_incident_workflow):

Field Label	Type	Reference / Choice
Number	Auto Number	Prefix INC, starts at 500, 5 digits
Caller	Reference	sys_user
Category	Choice	Network, Hardware, Access, Performance
Subcategory	Choice	Wi-Fi, Projector, Forgot Password, Slow Computer
Short Description	String	-
Description	String	-
State	Choice	New, In progress, On hold, Resolved, Closed
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user

3.3 Data Flow Diagram

Data moves through the system in the following sequence:

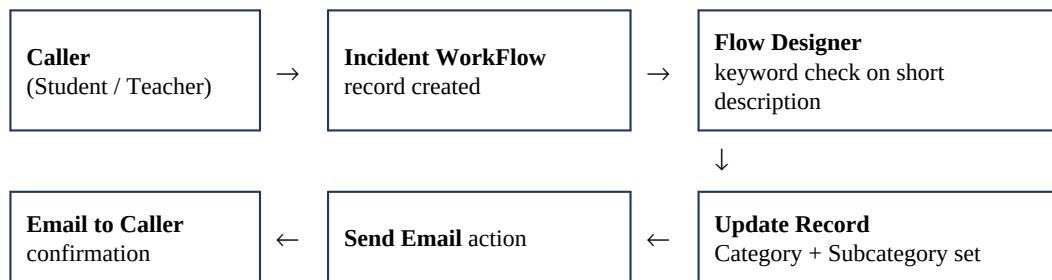


Figure 3.1 - Data flow of the auto classification solution

3.4 Technology Stack

Component	Technology
Platform	ServiceNow (Global application scope)
Data storage	Custom table Incident WorkFlow (u_incident_workflow)
Automation	Flow Designer (Workflow Studio) - no-code
Notification	Flow Designer Send Email action (SMTP)
Packaging	Local Update Set exported as XML

4. Project Design

4.1 Problem Solution Fit

Problem	Solution
Manual review of every ticket is slow.	Flow classifies the ticket the moment it is created.
Human error in choosing category.	Fixed keyword rules give consistent results.
Callers do not know if the ticket was received.	Automatic confirmation email to the caller.
Inconsistent subcategory choices.	Dependent choice list ties each subcategory to its category.

4.2 Proposed Solution

A custom table stores the tickets. A flow named **Auto Classify School IT Tickets** triggers when a record is created and the Category is empty. It then evaluates the short description with If / Else If conditions and updates the record:

Short description contains	Category	Subcategory
Wi-Fi or Network	Network	Wi-Fi
Projector	Hardware	Projector
Forgot password	Access	Forgot Password
Slow computer	Performance	Slow Computer

After classification, a **Send Email** action notifies the caller with the subject "Your Request for the issue has been submitted."

4.3 Solution Architecture

- **Data layer:** Incident WorkFlow table with auto-numbering (INC prefix) and dependent choice fields.
- **Automation layer:** Flow Designer flow with a Record Created trigger (condition: Category is empty), four classification branches and one email action.
- **Communication layer:** Email sent to the Caller's email address, logged under System Logs > Emails.
- **Deployment layer:** All changes captured in the Project Update Set and exported to XML.

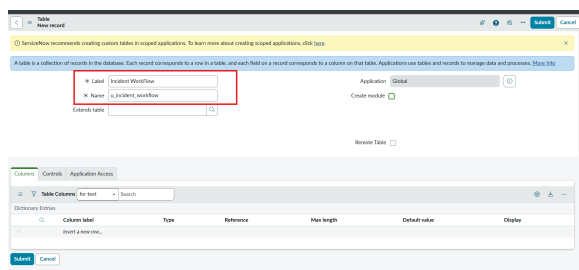


Figure 4.1 - Creating the Incident WorkFlow table

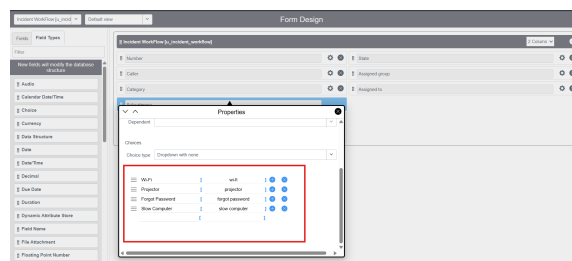


Figure 4.2 - Category choices in Form Design

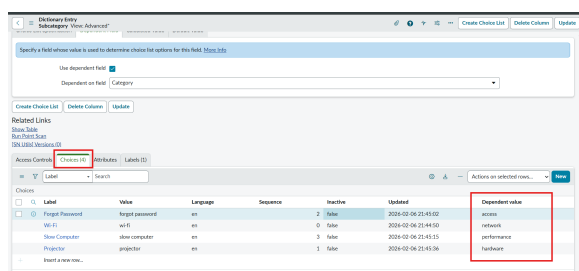


Figure 4.3 - Subcategory choices with dependent values

5. Project Planning & Scheduling

5.1 Project Planning

The project was carried out in phases, each with clear activities:

Phase	Activities
1. Requirement Analysis & Planning	Business use case; creation of the Project Update Set (Global scope).
2. Backend Development & Configuration	Custom table, fields and data types, Category / Subcategory dependency.
3. Automation	Flow Designer flow, trigger, four classification branches, email action, activation.
4. Testing, Validation & Security	Test scenarios, data validation checks, email log verification.
5. Deployment & Documentation	Update set set to Complete, exported to XML, final documentation.

6. Functional and Performance Testing

6.1 Performance Testing

Testing confirmed that the solution works as expected, meets the functional requirements, and follows basic security and access-control practice (the table's access controls are created for the u_incident_workflow_user role).

Data validation checks

- Mandatory fields are captured correctly.
- Auto-number is generated without duplication.
- Category and Subcategory values are stored accurately.
- Reference fields resolve correctly to the user and group tables.

Test	Input (Short description)	Expected result	Outcome
Scenario 1	Network-related issue (Wi-Fi / Network)	Category = Network, Subcategory = Wi-Fi; email sent	Passed
Scenario 2	Projector not turning on	Category = Hardware, Subcategory = Projector; email sent	Passed
Email check	Ticket created	Email record in System Logs with subject "Your Request for the issue has been submitted."	Passed

7. Results

7.1 Output Screenshots

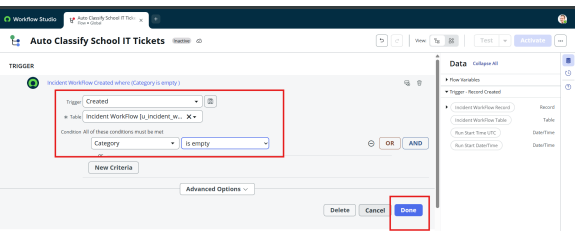


Figure 7.1 - Flow trigger: Record Created, Category is empty

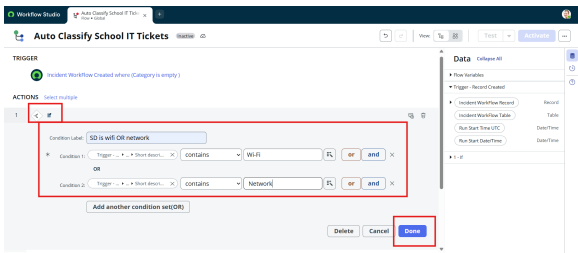


Figure 7.2 - Wi-Fi / Network condition

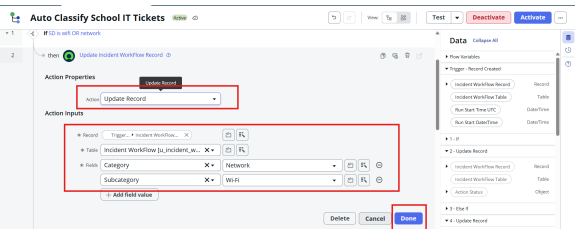


Figure 7.3 - Update Record sets Category and Subcategory

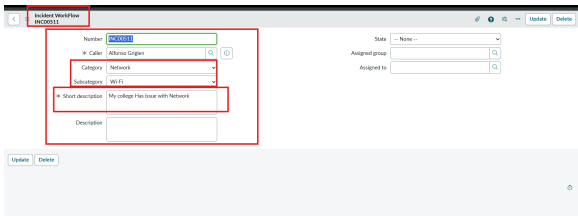


Figure 7.4 - INC00511 classified Network / Wi-Fi

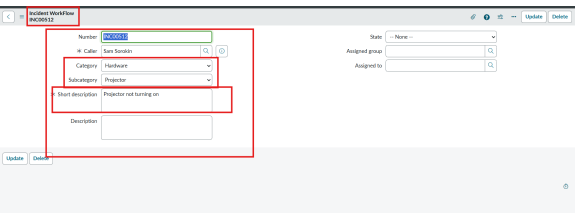


Figure 7.5 - INC00512 classified Hardware / Projector

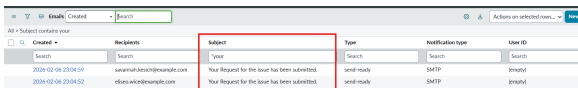


Figure 7.6 - Confirmation emails in System Logs

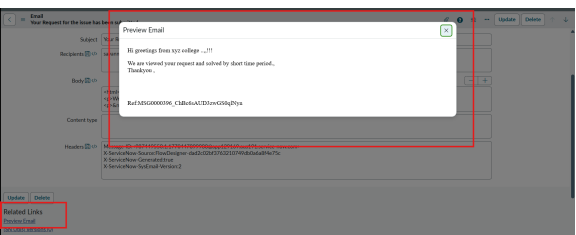


Figure 7.7 - Preview of the confirmation email

8. Advantages & Disadvantages

Advantages

- Consistent classification with no human error.
- Significantly less manual effort for IT staff.
- Callers are acknowledged instantly by email.
- No scripting required, so rules are easy to read, change and extend.
- Dependent choices keep category data clean and standardized.

Disadvantages

- Keyword matching is literal; spelling variations or unexpected wording (for example "WiFi" vs "Wi-Fi") may not match.
- Tickets that match no rule remain unclassified and need manual review.
- Only four issue types are covered at present.
- Rules are checked in order, so a description with several keywords takes the first match.

9. Conclusion

The Auto Ticket Classification using Flow Designer project delivers an end-to-end automation solution for a school IT helpdesk. Through structured requirement analysis, careful backend configuration and no-code automation, tickets are now classified automatically from the short description, and callers receive an instant confirmation email.

The phased approach kept the build clear and stable, and testing confirmed reliable behavior and data integrity. The design suits small and medium environments such as educational institutions and shows how ServiceNow can provide scalable, maintainable automation without complex scripting or machine learning.

10. Future Scope

- Automatic assignment of tickets to the right support group and agent.
- SLA tracking and escalation for unresolved tickets.
- Predictive Intelligence / machine-learning classification for free-text descriptions.
- A default fallback branch and broader keyword lists (for example "WiFi", "login", "hanging").
- Additional categories such as printers, software installation and email.

11. Appendix

Source Code

Not applicable - the solution is fully no-code. The configuration is captured in the exported update set (Project Update Set, state Complete, exported as XML).

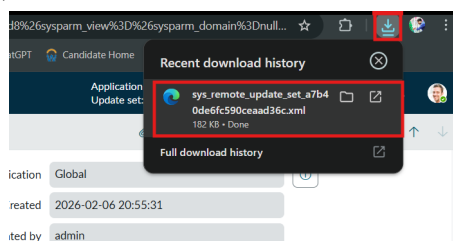


Figure A.1 - Update set exported to XML